



7.4 Lesson Plan: Windows Server Services

Class Objectives

By the end of the class, you will be able to:

- Understand Task Scheduling in a Windows environment
- Configure Windows Update on Windows Computers
- Understand the Windows Attack Space and which files and folders to watch as a Windows administrators

Lesson Slideshow

The lesson slides are available on Google Drive here: [7.4 Slides](#)

- **Note:** Editing access is not available for this document. If you or your students wish to modify the slides, please create a copy by navigating to File > "Make a copy..."

01. Welcome and Overview

Welcome to the fourth day of Windows. Lets review what we covered in the previous class:

- We used PowerShell to navigate and manage the Windows system directories and files.
- We secured our domain using group policy.
- We deployed Windows Defender to secure against malware threats.

Now that we have some familiarity with Active Directory, today we're going to learn how to manage **Windows Services and Files to Protect**

Accessing the Lab

Log in to your Windows RDP host machine and then start up and log in to the nested virtual machines if you have not done so already.

The Machines

Within our Windows RDP host machine, we have two nested Hyper-V virtual machines:

- A Windows 10 virtual machine that has Windows 10 installed on it just like our Windows RDP host machine.
- A Windows Server virtual machine that has the Windows Server 2019 operating system installed.

The Connectivity

Your Windows Azure Labs have already been set up with Active Directory Domain Services installed and configured.

The following are some details on how the machines are logically set up:

- Windows Server machine has Active Directory set up on it with the domain, `GoodCorp.net`. This will be the domain that we will be managing and installing ADCS on.
- The Windows 10 machine is connected to the Windows Server Active Directory domain—it is acting as a Windows client workstation that is set up so that it will receive and execute instructions and policies from the Active Directory server through a trust relationship.

02. Task Scheduling

In this section, you will use the administrative user, `Andrew`, created earlier in the module, to create scheduled tasks that will automate the reports we've been working on.

Task Scheduler is a GUI tool that allows system administrators to automate the execution of scripts and applications on a Windows system.

- These are similar to cron jobs in Linux. They can be set to execute at specific times or a certain amount of time after a user logs in.
- It is important to properly manage systems with scheduled tasks to automate security and system administration actions such as checking for updates for endpoint security software, sending important logs to systems such as SIEM (Security Information and Event Management), and scheduling system maintenance scripts.

Task Scheduling Demo

Continue using the Windows RDP host machine.

Explain the following demonstration scenario:

- Your CIO wants to schedule reports to be created on a daily basis. This system isn't remotely managed at the moment, and with a new developer coming in, huge changes might be made to the machine's configuration.
- This will help reduce workstation development drift for any new users for the Windows environment, and for this specific developer position and project.
- Your CIO has asked you to use the GUI tool Task Scheduler to create a task that will run each day.

Follow along on your machines.

- [Activity File: Task Scheduling](#)

Use the user, `azadmin`, to create a newly scheduled task.

- Click on the Windows search in the taskbar and search for "Task".
- Choose **Task Scheduler** from the list.
- Create a new task by selecting **Create Task...** on the **Actions** pane.
 - **Note:** If you don't see the Actions pane on the right, go to **View > Customize** and make sure Action Pane is enabled.
- Setting the Task Trigger
- In the Create Task GUI window, navigate to the **Triggers** tab, and click **New...**
- Set **Begin the task** to **On a schedule**.
- Make sure the task is set to repeat daily with **Recur every:[x] day** set to **1**, and the start date set to today. (It should already be set, but you may have to input it manually.)
- Finally, set the time to be a minute or two after the current time.
- Select **OK** to save your changes and return to the Create Task window.

03. Activity: Task Scheduling (0:15)

In this activity, you will be using **Task Manager** to create daily reports for the Chief Information Officer (CIO)

- [Activity File: Task Scheduling](#)

Answer any questions students may have.

04. Review: Task Scheduling

05. Windows Updates and Enterprise Management

Updates are the backbone of a secure and reliable IT environment, but they often go unnoticed in their day-to-day operations.

There are several reasons that patches and updates often go ignored:

- **Time:** Many systems are expected to be up continuously, but installing patches often requires rebooting of the servers.
- **Compatibility:** Many services on a Windows Server are highly-dependent on other services. Updating one service (i.e. SQL) could affect other services (i.e. PHP) downstream. Each service then has to be reconfigured to work with the updated services.
- **Broken Patches:** Even though patches are tested extensively, there are many variations of hardware and software that can cause issues with newly installed patches and the effects of these issues may not be identified until they cause problems in production systems.
- **Limited Resources:** Even if a patch is critical, if the system currently works, busy techs will delay installing the new updates in favor of maintaining a functional system.

Imagine this scenario: A critical vulnerability is discovered in Windows, leaving millions of devices worldwide susceptible to cyberattacks. Within days, Microsoft releases a security update to patch the issue. For a single user, enabling automatic updates might be sufficient. However, for an organization managing thousands of devices, a more robust and coordinated strategy is essential. This is where tools like SCCM (System Center Configuration Manager) and BCM (Business Continuity Management) come into play.

- **SCCM:** is a cornerstone of enterprise IT. It allows IT administrators to automate and customize the deployment of updates across an entire organization. Instead of relying on individual users, which can be inconsistent and prone to errors, to install updates, SCCM ensures that updates are systematically tested, approved, and distributed. For example, an administrator can schedule updates during non-peak hours, test patches on a subset of devices to avoid widespread issues, and monitor update compliance in real-time.
- **BCM:** on the other hand, focuses on maintaining operations during disruptions, including those caused by unplanned downtime during updates. By combining the capabilities of BCM with SCCM, organizations can minimize risks. For instance, backup protocols can ensure that critical data is protected before an update rollout, and fallback plans can address any adverse effects from a failed update.

Patch Tuesday:

One of the most critical elements of managing updates is “Patch Tuesday.” Occurring on the second Tuesday of each month, this is when Microsoft releases scheduled updates, including patches for vulnerabilities and performance improvements. Patch Tuesday provides predictability, allowing organizations to prepare in advance. IT teams can plan resource allocation, notify employees, and mitigate potential disruptions. However, Patch Tuesday also underscores the urgency of staying current—delaying updates leaves systems vulnerable to exploits, as attackers often target known vulnerabilities immediately after patches are released.

06. Demo: Configuring Windows Updates

In this demo, we will dive into the practical side of update management using Windows Server 2019. We'll explore how to configure automatic updates, manage update settings, and discuss enterprise tools like SCCM for efficient patch management. By following these steps, you'll gain hands-on experience with tools and settings that are vital for maintaining a secure and updated environment.

Access Windows Update Settings

1. Log in to the Windows Server 2019 machine with administrator privileges (sysadmin/p4ssw0rd*).
2. Open the Server Manager dashboard and navigate to Local Server.
3. Locate the Windows Update section in the Properties pane.
4. Click on the linked text under Windows Update, which will open the Windows Update settings.

Check for Updates

1. In the Windows Update window, click Check for updates to see if there are any pending updates.
2. Review the list of available updates, including security patches and feature updates.

Document the updates displayed for future reference and monitoring.

Step 3: Configure Automatic Updates

In the Windows Update settings, select Advanced options.

Choose the desired update settings:

Automatic (recommended): Ensures updates are downloaded and installed automatically during maintenance hours.

Notify to schedule restart: Provides flexibility for managing server reboots.

Define active hours to prevent automatic restarts during critical operation times.

Managing Updates with SCCM

SCCM can handle many of these functions automatically. **Note:** Configuring SCCM is beyond the scope of this module, provides successful deployment of patches.

1. Create a new Software Update Group to organize patches for deployment.
2. Select updates from the catalog that align with your organization's compliance policies.
 - Explain that compliance depends on the nature of the organization so it is important to understand the what the patch policies for the company are before deploying.
3. Deploy the update group to a test collection first to validate updates before broad deployment.
 - Explain to students that this isn't always practical, especially for zero-day attacks, but best practices are to deploy to a test environment first.
4. Schedule the deployment and monitor compliance using SCCM's reporting features.

07. Activity: Automatically Applying Patches

In this activity, you will be looking at the Windows Server 2019 to verify automatic update status and identifying which patches are currently not installed.

Activity file: [Activity File: Automatically Applying Patches](#)

08. Review: Automatically Applying Patches (0:10)

09. Break

10. Understanding Windows Attack Vectors

One of the greatest security risks in any Windows environment is the user space, a target-rich environment for attackers. Users, by necessity, often have access to critical files and systems, making their accounts a prime target for compromise. Attackers exploit this by blending into legitimate processes and services, hiding in common locations, and leveraging elevated privileges where possible.

Imagine an organization where a regular user has **local admin privileges**. A phishing email tricking them into running a malicious file could open the floodgates to a full-blown breach. Attackers frequently rely on user error or oversight, embedding their payloads in overlooked locations like temporary folders, scheduled tasks, or hidden AppData directories. These locations are less scrutinized in many environments, giving malicious files a foothold to operate undetected.

Malware developers are also adept at disguising their creations. They name their malicious services to blend in with Windows native services—like slightly altering names of legitimate processes such as `svchost.exe` to `svchost1.exe`. This could easily go unnoticed, allowing attackers to persist in the system for extended periods. This makes understanding native processes and services vital for identifying anomalies.

By building awareness of these tactics and vulnerabilities, you'll gain the ability to better secure systems and respond effectively to potential threats. This lesson empowers you to take control of user permissions, monitor suspicious activities, and strengthen the overall security posture of your systems.

11. Demo: User Space Attacks (0:30)

In this demonstration, we'll walk the students through a real-world scenario highlighting how attackers exploit user permissions, disguise their malware, and leverage common file locations for persistence. By examining practical tools and methods, you'll learn to identify and mitigate these threats effectively.

Imagine a scenario where a seemingly legitimate service is consuming unusually high CPU resources. An initial glance at the process name suggests it's a system service. However, a closer

inspection reveals discrepancies. With the right knowledge and tools, we can uncover this hidden threat and take steps to eliminate it.

Understand User Permissions

- Log in to the Windows 10 machine with administrator privileges.
- Open Computer Management and navigate to Local Users and Groups > Users.

Review user account permissions:

- Identify accounts with Local Administrator.
- Highlight accounts that should not have elevated access and discuss why reducing permissions is critical for security.

Inspect Common Malware Locations

Open File Explorer and navigate to common malware hiding spots:

- Click on `Local Disk (C:)`
- Review the following locations:
 - `C:\Windows\Tasks`
 - This directory stores scheduled tasks in Windows, which attackers can exploit to maintain persistence. By creating or modifying tasks, malware can execute at scheduled intervals without requiring user interaction, often going unnoticed by system administrators.
 - `C:\Windows\Temp`
 - The temporary files folder is frequently used by both legitimate applications and malware. Attackers can place malicious files here because it's a writable location accessible to many users and applications. It's often overlooked during routine security checks, making it an attractive hiding spot.
 - `C:\Users<username>\AppData`
 - This directory is specific to each user and stores application data and configurations. It's a common target because it's user-writable and hosts numerous subfolders where malicious files can blend in with legitimate data. Attackers can exploit this space to maintain persistence or run payloads that align with user privileges.

Examine Running Processes

- Open Task Manager and navigate to the Processes tab.
 - **Note:** You may have to click on the `More Details` button at the bottom.
- Sort processes by CPU or memory usage to identify anomalies.
- Explain to students that they can search the process name on the internet to get additional information on the process.

Review Windows Services

- Open the Services application by typing `services.msc` in the Run dialog.

- Double-click on the BitLocker Drive Encryption Service
- Demonstrate to students the:
 - Service name: BDESVC
 - Executable Path: C:\Windows\System32\svchost.exe -k netsvcs -pPoint out that the service uses `svchost.exe` as the executable.
Send students the link [svchost.exe](#) for additional information on `svchost.exe`.
- Explain how attackers disguise malicious services by mimicking native service names. For example:
 - A legitimate service like `svchost.exe` may be spoofed with a name like `svchost1.exe`.
 - Cross-check the service name and description with official documentation.
 - Review the service's executable path for anomalies.

12. Activity: Investigating User Spaces and Services (0:20):

In this activity, you will act as a system administrator securing a corporate network.

[Activity File: Investigating User Space and Services](#)

13. Review

14. End

Congratulations on completing the Windows Administration Module.